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13 June 1958
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MILITARY SPECIFICATION

DESICCATOR UNIT FOR FIRE CONTROL INSTRUMENTS

1. SCOPE

1.1 Scope.- This specification covers desiccating units for use on various fire control instruments.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-B-676	Boxes, Set-up, Paperboard
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant

Military

MIL-P-116	Preservation, Methods of
MIL-B-138	Boxes, Wood, Fiberboard-lined for Overseas Shipment (For Weight of Contents Not Exceeding 500 pounds)

FSC 4440

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MIL-D-3716	Desiccants (Activated) for Dynamic Dehumidification
MIL-P-14118	Plastic, Cellulose-Acetate, Sheets and Film

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- The unit shall be made of transparent plastic material conforming to Specification MIL-P-14118.

3.1.1 Dehydrating agent.- The dehydrating agent shall be Silica-Gel, Type IV, Grade H, in accordance with Specification MIL-D-3716.

3.2 Size and dimensions.- The size and dimensions of the unit shall conform to those shown on the drawings specified in the contract or purchase order.

3.3 Temperature.- The unit shall not deform in any manner and shall not show any visible defects when subjected to the temperature extremes of -80°F (-62.2°C) and 160°F (71°C).

3.4 Water-vapor transmission (seal).- The gross weight of a loaded and sealed desiccator unit shall not increase more than three (3) percent when exposed to any temperature between 95°F (35°C) and 105 F (40.5 C) inclusive and a relative humidity of 95 percent or more for a period of twenty-four (24) hours.

3.5 Adsorption rate of unsealed unit.- An unsealed unit shall adsorb a minimum of 1.5 grams of moisture per square inch (including the area of perforations) of the perforated end per 24 hour period when subjected to any temperature between 95°F (35°C) and 105°F (40.5°C) inclusive and a relative humidity of 95 percent or more.

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3.6 Retention of dehydrating agent with seal removed.- There shall be no evidence of silica-gel leakage when dropped a specified number of times perforated side down.

3.7 Workmanship.- The unit shall be free from mold marks, cracks, splits and other surface or internal defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.

4.1.1 Contractor inspection.- Unless otherwise specified herein, the inspections required by this specification will be performed by the contractor.

4.1.2 Contractor quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. Results of all examinations and tests performed under the quality assurance system shall be made available to the Government. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the specification and other documents referenced in this specification. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that the substitute procedures will provide, as a minimum, the quality assurance required in the contractual documents. In cases of dispute as to whether certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced therein.

4.1.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of poor practices which might have an affect upon the quality of the product will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid

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interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Inspection provisions.

4.2.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.2.2 Examination.- The classification of defects in Tables I, II and III shall constitute the minimum inspection requirements to be performed prior to acceptance.

4.2.2.1 Disposition.- All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS
(Use Inspection Level I, and Sampling Plan Table IV-A of Standard MIL-STD-105)

Major	AQL - 4.0%	Req. Par.	QA Par.	Test Method
1.	Material	3.1	4.3.1	Certification
2.	Dehydrating agent	3.1.1	4.3.1.1	Certification
3.	Size and dimensions	3.2	4.3.2	Standard Measuring Equipment (SME)
4.	Temperature	3.3	4.3.3	Cold Chest and Oven
5.	Water-vapor transmission	3.4	4.3.4	SME
6.	Adsorption rate	3.5	4.3.5	SME
7.	Retention of dehydrating agent	3.6	4.3.6	SME
8.	Workmanship	3.7	4.3.7	Visual

4.2.3 Inspection equipment.- Unless otherwise specified in the contract, the contractor shall provide and maintain gages and other measuring and testing devices considered necessary for inspection of supplies for conformance to contract requirements. The Government reserves the right to certify all such equipment and to use this equipment for Government verification inspection.

4.3 Test methods and procedures.

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4.3.1 Material.- The contractor shall furnish the Government with certification that material used in the fabrication of the unit shall conform to the requirements in 3.1.

4.3.1.1 Dehydrating agent.- The contractor shall furnish the Government with certification that the dehydrating agent used shall conform to the requirements of 3.1.1.

4.3.2 Size and dimensions.- The unit shall be inspected by means of standard measuring equipment (SME) for conformance to the requirements referenced in the applicable drawings.

4.3.3 Temperature.- The unit shall be placed in the test chamber and the temperature reduced gradually to minus 80°F (-62.2°C). The unit shall remain at this temperature until thermal stabilization is reached. The temperature shall then be raised gradually to 160°F (71°C). The unit shall remain in the chamber until thermal stabilization is reached. The temperature shall then be reduced to standard ambient temperature at which time the unit shall be tested to meet the requirements of 3.3.

4.3.4 Water-vapor transmission (seal).- A sealed unit shall be weighed and subjected to the conditions described in 3.4, using equipment or equal described in Specification MIL-D-3716. At the end of the stated time, the unit shall be superficially dried and reweighed. The percentage of moisture adsorption shall be computed by the following formula:

$$\frac{A-B}{B} \times 100 = \% \text{ Moisture Adsorption}$$

where B = initial gross weight of unit
A = final gross weight of unit

The gross weight of a loaded and sealed desiccator unit shall not increase more than three (3) percent when exposed to any temperature between 95°F (35°C) and 105°F (40.5°C) inclusive and a relative humidity of 95 percent or more for a period of twenty-four (24) hours.

4.3.5 Adsorption rate of unsealed unit.- An unsealed unit shall adsorb a minimum of 1.5 grams of moisture per square inch (including the area of perforations) of the perforated end per twenty-four (24) hour period when subjected to any temperature between 95°F (35°C) and 105°F (40°C) inclusive and a relative humidity of 95 percent or more. A unit with the seal removed shall be subjected to the conditions specified herein from which the adsorption rate shall be calculated.

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4.3.6 Retention of dehydrating agent.- The unit with the seal removed shall be dropped ten (10) times, perforated side down, from a height of three (3) inches on a polished surface. The polished surface shall then be examined under a 3 power lens for evidence of particles of silica-gel. In cases where the unit is used in an optical instrument, the polished surface shall be examined with a lens of the same power of the eyepiece of the instrument.

4.3.7 Workmanship.- Workmanship shall be visually inspected as specified in 3.5.

TABLE II - CLASSIFICATION OF DEFECTS
(Use Inspection Level L1, and Sampling Plan Table IV-A of Standard MIL-STD-105)

Major AQL - 2.5%	Req. Par.	Test Method
1. Markings illegible, missing (Levels A, B, and C)	5.3	Visual
2. Water-vaporproof bag, damaged or improperly sealed (Level A)	5.1.1.3	Visual
3. Improper cleaning (Level A)	5.1.1.1	Visual
4. Improper drying (Level A)	5.1.1.2	Visual
Minor AQL - 4.0%		
41. Unit quantity not as specified (Level A)	5.1.1.3	Visual
42. Intermediate quantities not as specified (Level A)	5.1.1.4	Visual
43. Defective taping or sealing of paperboard box	5.1.1.3	Visual
44. Packing (Levels A, B, and C)	5.2	Visual
45. Improper strapping (Level A)	5.2.1	Visual

TABLE III - CLASSIFICATION OF DEFECTS
(Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105)

Major AQL - 2.5%	Req. Par.	QA Par.	Test Method
Heat Seal (Level A)	5.1.1.3	4.4.1	Visual
Quick Leak (Level A)	5.1.1.3	4.4.1	Visual

4.4 Test methods and procedures.

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4.4.1 Packaging tests.- Each package in the applicable sample shall be subjected to the pertinent packaging tests of Specification MIL-P-116.

4.4.2 Determination of cleanliness test.- Periodically the applicable surfaces of one or more units shall be subjected to the "Determination of Cleanliness" test in accordance with Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning.- Cleaning shall be accomplished to remove any deposit foreign to the composition of the item in accordance with process C-1 of Specification MIL-P-116.

5.1.1.2 Drying.- The desiccators shall be dried in accordance with any drying procedure listed in Specification MIL-P-116.

5.1.1.3 Unit packaging.- Each desiccator unit shall be packaged method IA-15 in accordance with Specification MIL-P-116 using a paperboard folding or set-up box conforming to the requirements of Specification PPP-B-566 and PPP-B-676, respectively. Desiccators with stems shall be protected with die-cut pads fabricated from material meeting the requirements of Specification PPP-B-636, class 1.

5.1.1.4 Intermediate packaging.- Twenty identical desiccator units (desiccator units of the same stock number) packaged as specified in 5.1.1.3 shall be further packaged in a close-fitting intermediate container conforming to the requirements of Specification PPP-B-636, class 2, W5s. Seams, joints and closures of intermediate containers shall be taped with material measuring 2 inches in width conforming to the requirements of Specification PPP-T-76.

5.1.2 Level C.

5.1.2.1 Cleaning and drying.- Cleaning and drying shall be in accordance with the requirements as specified in 5.1.1.1 and 5.1.1.2.

5.1.2.2 Unit packaging.- Each desiccator unit shall be packaged method III in accordance with Specification MIL-P-116 using a box as specified in 5.1.1.3.

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5.2 Packing.

5.2.1 Level A.- Identical desiccators (desiccator units of the same stock number) packaged as specified in 5.1.1.4 shall be packed in a wirebound wood box, wood-cleated fiberboard box, wood cleated-plywood box, nailed wood box, fiberboard box or a fiberboard-lined wood box conforming to the requirements of Specification PPP-B-585 class 3, PPP-B-591 (Overseas type), PPP-B-601 style A or B, type I or II, class 2, PPP-B-621 class 2, PPP-B-636 class 2, and MIL-B-138, respectively. Closure and strapping requirements shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed approximately 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 Level B.- Identical desiccators (desiccator units of the same stock number) packaged as specified in 5.1.1.4 shall be packed in a wirebound wood box, wood-cleated fiberboard box, wood cleated-plywood box, nailed wood box or a fiberboard box conforming to the requirements of Specification PPP-B-585 class 1, PPP-B-591 (Domestic type), PPP-B-601 (Domestic type), PPP-B-621 class 1 and PPP-B-636 class 1, respectively. Closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed approximately 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 Level C.- Identical desiccators (desiccator units of the same stock number) packaged as specified in 5.1.2.2 shall be packed in a wirebound wood box, wood-cleated fiberboard box, wood cleated-plywood box, nailed wood box or a fiberboard box conforming to Specification PPP-B-585 class 1, PPP-B-591 (Domestic type), PPP-B-601 (Domestic type), PPP-B-621 class 1 and PPP-B-636 class 1, respectively. Closure shall be as specified in the applicable box specification or appendix thereto.

5.3 Marking.- Unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Desiccating units are intended for use in removing moisture from fire control instruments while in service.

6.2 Ordering data.- Procurement documents should specify the following:

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a. Title, number and date of this specification.
b. Drawing of unit assembly.
c. Quantity required.
d. Selection of applicable level of packaging and packing.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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